National University of Computer and Emerging Sciences
Islamabad Campus

# Metal Slug

## Object Oriented Programming Project Rubric
## (Spring 2025) [1000+250]

**Object Oriented Programming Project Self-Evaluation**

**Student 1 Name: ____________________________**

**Roll Number: ____________________________**

**Section: ___**

**Student 2 Name: ____________________________**

**Roll Number: ____________________________**

**Section: ___**

**Student 3 Name: ____________________________**

**Roll Number: ____________________________**

**Section: ___**

**Object Oriented Programming Concepts (110 Marks)**

| Sr # # | Feature | Total Marks | Self Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 1 | Proper use of classes, objects, and core OOP principles including encapsulation, inheritance, and polymorphism. Code must demonstrate appropriate use of access modifiers, data hiding, and | 40 | | |

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| | clear object interactions. Note: Apply good programming practices | | | |
| 2 | Viva Questions from principles of Object Oriented Programming | 70 | | |

**Core Game Mechanics (70 Marks)**

| Sr # | Feature | Total Marks | Self Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 3 | Game mode selection screen i.e. Survival mode and campaign mode | 3 | | |
| 4 | Base weapons implementation: A pistol with infinite bullets and a knife for close range | 5 | | |
| 5 | Shooting mechanism deals accurate damage to the enemies | 5 | | |
| 6 | Proper animations for shooting and projectiles | 5 | | |
| 7 | Accurate aiming mechanism as defined in the project statement | 5 | | |
| 8 | Grenades implementation and trajectory logic as defined in the statement | 7 | | |
| 9 | Proper animations for grenade projectiles with | 5 | | |

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|---|---|---|---|---|
| | ballistic trajectories | | | |
| 10 | Switching between 4 characters upon pressing the ‘Z’ key. | 10 | | |
| 11 | Accurate depiction of state changes upon enemy attacks on players.. State changes include damage states and the undead and mummy states | 10 | | |
| 12 | Accurate implementation of fusion companion following OOP principles. | 15 | | |

**Enemy System (90 Marks)**

| Sr # | Feature | Total Marks | Self Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 13 | Rebel Soldier | 4 | | |
| 14 | Rebel Soldier Animations | 3 | | |
| 15 | Shielded Soldier | 6 | | |
| 16 | Shielded Soldier Animations | 4 | | |
| 17 | Bazooka Soldier | 7 | | |
| 18 | Bazooka Soldier Animations | 4 | | |
| 19 | Grenade Soldier | 7 | | |
| 20 | Grenade Soldier | 4 | | |
| 21 | Paratroopers | 12 | | |

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|---|---|---|---|---|
| 22 | Paratroopers Animations | 3 | | |
| 23 | Mummy Warrior | 8 | | |
| 24 | Mummy Warrior Animations | 4 | | |
| 25 | Zombie | 8 | | |
| 26 | Zombie Animations | 4 | | |
| 27 | Martian | 8 | | |
| 28 | Martians Animations | 4 | | |

**Vehicle System (65 Marks)**

| Sr # | Feature | Total Marks | Self Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 29 | Flying Tara | 5 | | |
| 30 | Flying Tara Animations | 3 | | |
| 31 | M15-A Bradley | 5 | | |
| 32 | M15-A Bradley Animation | 3 | | |
| 33 | Enemy Sub | 5 | | |
| 34 | Enemy Sub Animation | 3 | | |
| 35 | Metal Slug | 5 | | |
| 36 | Metal Slug Animation | 3 | | |

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|---|---|---|---|---|
| 37 | Slug Flyer | 5 | | |
| 38 | Slug Flyer Animation | 3 | | |
| 39 | Slug Mariner | 5 | | |
| 40 | Slug Mariner Animation | 3 | | |
| 41 | Amphibious Slug | 10 | | |
| 42 | Amphibious Slug Animation | 7 | | |

**Weapon System (50 marks)**

| Sr # | Feature | Total Marks | Self Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 43 | Heavy Machine Gun | 5 | | |
| 44 | Rocket Launcher | 5 | | |
| 45 | Flame Shot | 10 | | |
| 46 | Laser Gun | 5 | | |
| 47 | Hand Grenade | 5 | | |

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|---|---|---|---|---|
| 48 | Fire Bomb Grenade | 10 | | |
| 49 | Weapon Animation | 10 | | |

**Level Design & Gamemode Design (130 Marks) 13%**

| Sr # | Feature | Total Marks | Self Assigned Marks | Evaluation Marks |
|---|---|---|---|---|
| 50 | 3 Survival Levels with all three biomes and accurate enemy spawning | 40 | | |
| 51 | 1 Campaign mode level with accurate biomes and enemies with infinite map generation leveraging fractal / perlin noise | 90 | | |

**Collectibles (30 Marks)**

| Sr # | Feature | Total Marks | Self Assigned Marks | Evaluation Marks |
|---|---|---|---|---|
| 52 | Food with saturation replenishing | 4 | | |
| 53 | POW prisoner with crate dropping | 13 | | |
| 54 | Supply Crate | 13 | | |

**Boss and Boss Level (125 Marks)**

| Sr # | Feature | Total Marks | Self Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 55 | Iron Nokana | 10 | | |

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|---|---|---|---|---|
| 56 | Iron Nokana Animations | 10 | | |
| 57 | Hairbuster Riberts | 10 | | |
| 58 | Hairbuster Riberts Animations | 10 | | |
| 59 | Sea Satan | 10 | | |
| 60 | Sea Satan Animation | 10 | | |
| 61 | Ultimate Enemy | 20 | | |
| 62 | Minion spawning in accurate batches | 10 | | |
| 63 | Phase 1 level specs | 5 | | |
| 64 | Phase 2 level specs | 5 | | |
| 65 | Phase 3 level specs | 5 | | |
| 66 | Phase 4 collision, fusion and specs | 20 | | |

**Scoring System (30 Marks)**

| Sr # | Feature | Total Marks | Self Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 67 | Enemy and Boss Kills | 5 | | |
| 68 | Bonuses on Kills | 5 | | |
| 69 | Level clear scores | 10 | | |

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| 70 | Highscore leaderboard in main menu | 10 | | |

**Special Features (300 Marks)**

| Sr # | Feature | Total Marks | Self Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 71 | Fannum Tax (animation) | 15 | | |
| 72 | The Art of self playing | 75 | | |
| 73 | Developer Mode | 30 | | |
| 74 | Forks of Fate | 25 | | |
| 75 | The Reflex Fingerprint | 30 | | |
| 76 | Memory of Cowardice | 15 | | |
| 77 | Echo Chambers | 15 | | |
| 78 | Urdu Assembly injection | 15 | | |
| 79 | Chains of Death | 50 | | |
| 80 | The Fingerprint file and Ghost | 30 | | |

**Bonus Animations (100 Marks)**

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| Sr # | Feature | Total Marks | Self Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 81 | **Visual Fluidity & Easing:** Animations are smooth without jarring jumps, clipping, or robotic linear movement. | 30 | | |
| 82 | **Logic & Frame-Rate:** Animation speed is consistent across different hardware (uses delta time/fixed timestep) and handles states correctly. | 30 | | |
| 83 | **Performance & Memory:** Code runs efficiently without memory leaks, massive frame drops. | 20 | | |
| 84 | **Interactivity:** Animations respond accurately and immediately to input, and can transition cleanly between states. | 20 | | |

## Bonus Features (150 Marks)

| Sr # | Feature | Total Marks | Self Assigned Marks | Evaluated Marks |
|---|---|---|---|---|
| 85 | Grudgful Enemies | 20 | | |
| 86 | The Ragebait | 25 | | |
| 87 | The Death Compositor | 25 | | |

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|---|---|---|---|---|
| 88 | Voice of the Bene Gesserit | 50 | | |

### Penalties (To Be Filled by by the evaluators during the demo) (-∞ Marks)

| Code | Violation Category | Description of Offense | Max Penalty | Penalty Applied |
|---|---|---|---|---|
| **P1** | **Anti-Polymorphism** | Using switch statements, if/else chains for type-checking, or manual RTTI (typeid, dynamic_cast abuse) instead of relying on dynamic dispatch, virtual functions, and abstract base classes. | **-100** | |
| **P2** | **UML & Architecture Diversion** | Unjustified deviation from the provided UML diagram. Includes missing required classes, inventing unnecessary classes, or implementing incorrect relationships (e.g., using Composition instead of Aggregation, or breaking IS-A relationships). | **-50** | |
| **P3** | **Memory & Lifetime Mismanagement** | Failure to manage dynamic memory correctly. Includes memory leaks, missing virtual destructors in polymorphic base classes, or failing to implement deep copies (violating the Rule of Three/Five). | **-150** | |

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|---|---|---|---|---|
| **P4** | **Encapsulation Breach** | Exposing state via public data members. Using superficial getters/setters that merely expose internal representations without enforcing class invariants or data hiding. | **-50** | |
| **P5** | **Design Pattern Violation** | Ignoring explicitly requested design patterns, or forcing a pattern where it does not fit (e.g., shoehorning a Singleton, or implementing a Factory that is essentially just a massive switch statement). | **-60** | |
| **P6** | **Tight Coupling** | Classes are overly dependent on the concrete implementations of other classes rather than interfaces/abstract classes, creating rigid code that is difficult to extend. | **-40** | |
| **P7** | **Plagiarism** | A group caught with plagiarised code. | **-∞** | |
| **P8** | **AI Usage** | Any group found to have done portions of code or core implementation using AI be it even formatting shall be dealt strictly and may result in the nullification of the feature set or the whole project | **-30 to -1000** | |